

1.7 LINEAR INDEPENDENCE

An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution. The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** if there exist weights $c_1, \dots, c_p$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0} \quad (2)$$

$c_1, c_2, \dots, c_p$ - weights
 $\rightarrow$ lin. dep. relation

EXAMPLE 1 Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$.

- Determine if the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.
- If possible, find a linear dependence relation among $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$.

$$\left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 0 & 0 & 0 \\ 2 & 5 & 1 & 0 & 0 & 0 \\ 3 & 6 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 0 & 0 & 0 \\ 0 & -3 & -3 & 0 & 0 & 0 \\ 0 & -6 & -6 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{cases} x_1 - 2x_3 = 0 \\ x_2 + x_3 = 0 \end{cases} \rightarrow \begin{cases} x_1 = 2x_3 \\ x_2 = -x_3 \end{cases} \quad \begin{array}{l} x_3 - \text{free} \\ x_1, x_2 - \text{basic} \end{array}$$

$$x_3 = 2 \Rightarrow x_2 = -2, \quad x_1 = 4$$

$$c_1 = 4, \quad c_2 = -2, \quad c_3 = 2 \quad - \text{weights}$$

$$\boxed{4\mathbf{v}_1 - 2\mathbf{v}_2 + 2\mathbf{v}_3 = \mathbf{0}} \quad \text{lin. dep. rel.}$$

Linear Independence of Matrix Columns

Suppose that we begin with a matrix $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ instead of a set of vectors. The matrix equation $A\mathbf{x} = \mathbf{0}$ can be written as

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$$

Each linear dependence relation among the columns of A corresponds to a nontrivial solution of $A\mathbf{x} = \mathbf{0}$. Thus we have the following important fact.

The columns of a matrix A are linearly independent if and only if the equation $A\mathbf{x} = \mathbf{0}$ has *only* the trivial solution. (3)

EXAMPLE 2 Determine if the columns of the matrix $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly independent.

SOLUTION To study $A\mathbf{x} = \mathbf{0}$, row reduce the augmented matrix:

$$\left[\begin{array}{cccc} 0 & 1 & 4 & 0 \\ 1 & 2 & -1 & 0 \\ 5 & 8 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cccc} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & -2 & 5 & 0 \end{array} \right] \sim \left[\begin{array}{cccc} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 13 & 0 \end{array} \right]$$

At this point, it is clear that there are three basic variables and no free variables. So the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, and the columns of A are linearly independent. ■

$$\begin{cases} x_1 + 2x_2 - x_3 = 0 \\ x_2 + 4x_3 = 0 \\ 13x_3 = 0 \end{cases} \Rightarrow \begin{aligned} & x_1 = 0 \\ & x_3 + 4 \cdot 0 = 0 \Rightarrow x_3 = 0 \\ & \Rightarrow x_2 = 0 \end{aligned}$$

Only trivial solution

Indep

Theorem

If the set contains vectors more than there are entries in each vector then this set is lin. dep

Proof: Let it be a set with p -vectors
Each vector has no more than n -entr.

n rows $\left[\begin{array}{c} \text{number of variables} \\ \text{is larger than equal} \\ \text{to } p \end{array} \right] p > n$
 p columns

there can be only n basic variables
 $p-n$ - variables free inf many solns.
not trivial

$\Downarrow$
dep

Theorem

If the set contains zero vector then it is dep

Proof: $\vec{v}_1 \quad \vec{v}_2 \quad \vec{v}_3 \quad \dots \quad \vec{v}_p \quad \vec{0}$

$$0 \cdot \vec{v}_1 + 0 \cdot \vec{v}_2 + \dots + 0 \cdot \vec{v}_p + 5 \cdot \vec{0} = \vec{0} \quad \checkmark$$

1.8 INTRODUCTION TO LINEAR TRANSFORMATIONS

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 4 & 0 & 5 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \cdot 1 - 3 \cdot 1 + 1 \cdot 1 + 3 \cdot 1 \\ 4 \cdot 1 + 0 \cdot 1 + 5 \cdot 1 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

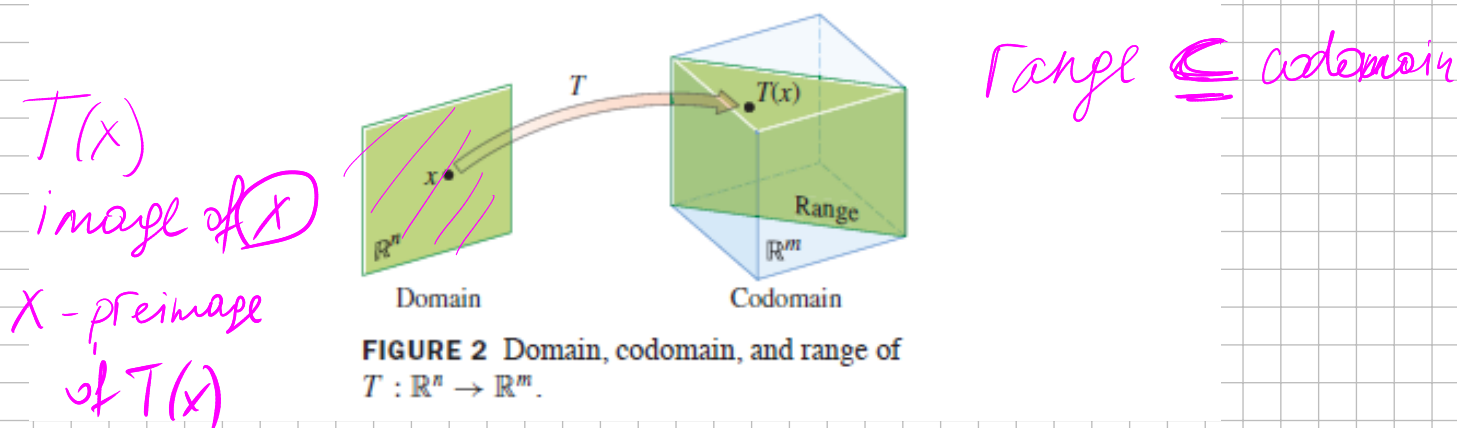
$\downarrow A_{4 \times 2}$ $\downarrow \mathbb{R}^4$ $\downarrow \mathbb{R}^2$

$$A_{m \times n} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 0 \end{bmatrix} \rightarrow \text{projection onto } Oxy\text{-plane}$$

$A: T(x): \mathbb{R}^3 \rightarrow \mathbb{R}^3$

A transformation (or function or mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$. The set $\mathbb{R}^n$ is called the **domain** of T , and $\mathbb{R}^m$ is called the **codomain** of T . The notation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$. For $\mathbf{x}$ in $\mathbb{R}^n$, the vector $T(\mathbf{x})$ in $\mathbb{R}^m$ is called the **image** of $\mathbf{x}$ (under the action of T). The set of all images $T(\mathbf{x})$ is called the **range** of T . See Figure 2.



EXAMPLE 1 Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$, and define a transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

- Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T . $\begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$
- Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$. $\begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$
- Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$? no
- Determine if $\mathbf{c}$ is in the range of the transformation T . no

a) $T(\mathbf{u}) = A \cdot \mathbf{u} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$

b) $A \cdot \mathbf{x} = \mathbf{b} \Rightarrow \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} \Rightarrow \begin{cases} x_1 - 3x_2 = 3 \\ 3x_1 + 5x_2 = 2 \\ -x_1 + 7x_2 = -5 \end{cases}$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 1 & -1/2 \end{array} \right] \sim$$

$$\sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{array} \right] \Rightarrow \begin{cases} x_1 - 3x_2 = 3 \\ x_2 = -1/2 \end{cases} \Rightarrow \begin{cases} x_1 = 3(1 - \frac{1}{2}) = \frac{3}{2} \\ x_2 = -1/2 \end{cases}$$

$$\mathbf{x} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix} \text{ - preimage of } \mathbf{b}$$

c) no, $\mathbf{x}$ is unique

d) IS the system $A\mathbf{x} = \mathbf{c}$ consistent?

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 1 & 2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & -1/2 \\ 0 & 0 & 5/2 \end{array} \right]$$

$\left[\begin{array}{cc|c} 0 & 0 & 5/2 \end{array} \right]$ no solution - $\mathbf{c}$ is not from the range

A transformation (or mapping) T is **linear** if:

- (i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T ;
- (ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T .

- (i) Image of sum is equal to sum of images
- (ii) Image of scaling is equal to scaling of the image.

$$T(\mathbf{x}) = A \cdot \mathbf{x} \quad A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}; \quad \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$A \cdot (\mathbf{u} + \mathbf{v}) = A \cdot \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 + 2u_2 + 2v_2 \\ 3u_1 + 3v_1 + 4u_2 + 4v_2 \end{bmatrix}$$

$$A\mathbf{u} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} u_1 + 2u_2 \\ 3u_1 + 4u_2 \end{bmatrix} \quad \left| \quad A\mathbf{v} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 + 2v_2 \\ 3v_1 + 4v_2 \end{bmatrix}\right.$$

$$A\mathbf{u} + A\mathbf{v} = \begin{bmatrix} u_1 + 2u_2 + v_1 + 2v_2 \\ 3u_1 + 4u_2 + 3v_1 + 4v_2 \end{bmatrix} = A(\mathbf{u} + \mathbf{v})$$

EXAMPLE 4 Given a scalar r , define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = r\mathbf{x}$. T is called a **contraction** when $0 \leq r \leq 1$ and a **dilation** when $r > 1$. Let $r = 3$, and show that T is a linear transformation.

$$T(\mathbf{x}) = 3 \cdot \mathbf{x}$$

$$\underline{T(\mathbf{x} + \mathbf{y})} = 3 \cdot (\mathbf{x} + \mathbf{y}) = 3\mathbf{x} + 3\mathbf{y} = T(\mathbf{x}) + T(\mathbf{y})$$

$$\underline{T(c \cdot \mathbf{x})} = 3 \cdot c\mathbf{x} = c \cdot (3\mathbf{x}) = \underline{c \cdot T(\mathbf{x})}$$

1.9 THE MATRIX OF A LINEAR TRANSFORMATION

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

In fact, A is the $m \times n$ matrix whose j th column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j th column of the identity matrix in $\mathbb{R}^n$:

$$A = [T(\mathbf{e}_1) \quad \cdots \quad T(\mathbf{e}_n)] \quad (3)$$

EXAMPLE 2 Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for $\mathbf{x}$ in $\mathbb{R}^2$.

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{T(\mathbf{e}_1)} = 3 \cdot \mathbf{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad \underline{T(\mathbf{e}_2)} = 3 \cdot \mathbf{e}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$T(\mathbf{x}) = A\mathbf{x} \Rightarrow A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2)]$$

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\mathbb{R}^2 \quad \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathbf{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

$$\mathbb{R}^3 \quad \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

TABLE 1 Reflections

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Reflection through the x_2 -axis		$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Reflection through the line $x_2 = x_1$		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Reflection through the line $x_2 = -x_1$		$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
Reflection through the origin		$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

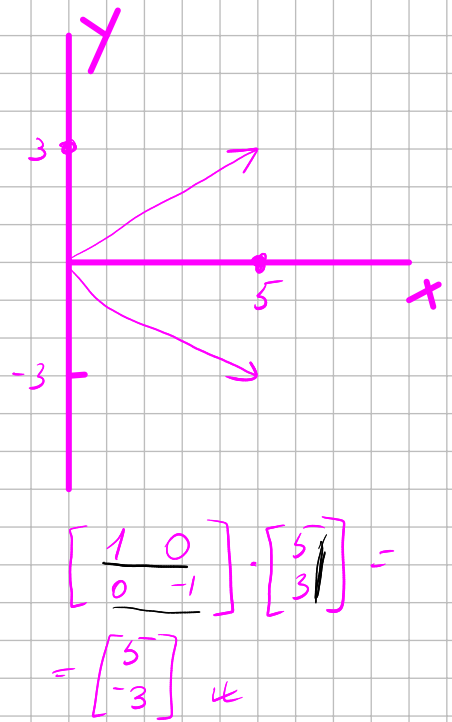


TABLE 2 Contractions and Expansions

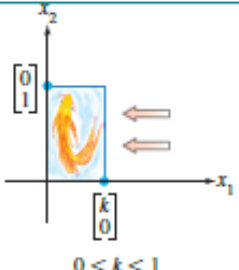
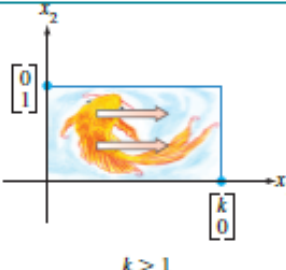
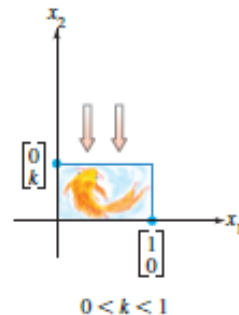
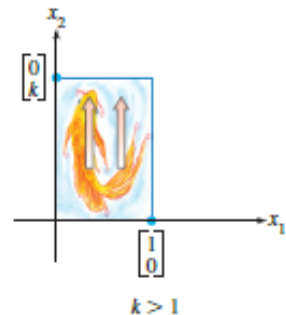
Transformation	Image of the Unit Square		Standard Matrix
Horizontal contraction and expansion			$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$
	$0 < k < 1$	$k > 1$	
Vertical contraction and expansion			$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$
	$0 < k < 1$	$k > 1$	

TABLE 3 Shears

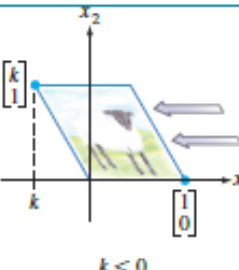
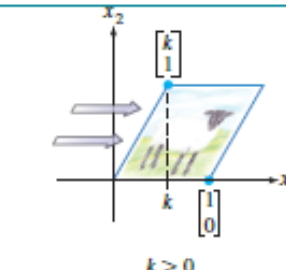
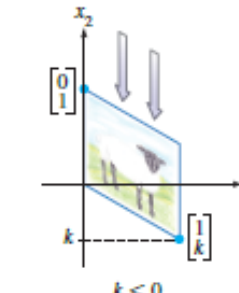
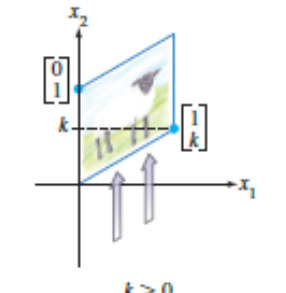
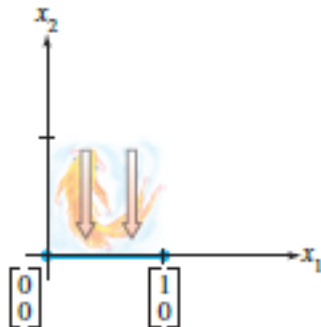
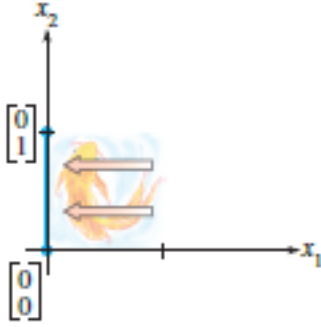
Transformation	Image of the Unit Square		Standard Matrix
Horizontal shear			$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
	$k < 0$	$k > 0$	
Vertical shear			$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$
	$k < 0$	$k > 0$	

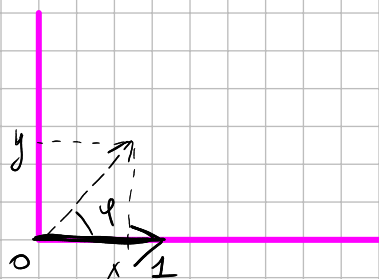
TABLE 4 Projections

Transformation	Image of the Unit Square	Standard Matrix
Projection onto the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
Projection onto the x_2 -axis		$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

EXAMPLE 3 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle φ , with counterclockwise rotation for a positive angle. We could show geometrically that such a transformation is linear. (See Figure 6 in Section 1.8.) Find the standard matrix A of this transformation.

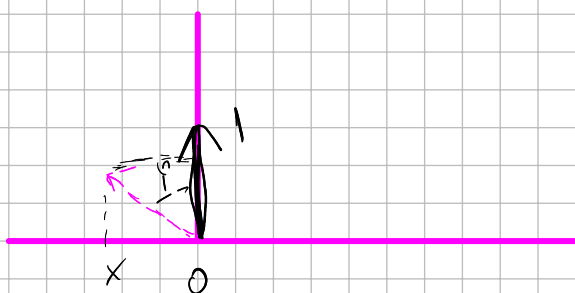
$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$T(e_1) = \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}$$



$$\begin{aligned} \sin \varphi &= y \\ \cos \varphi &= x \end{aligned}$$

$$e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$\begin{aligned} x &= -\sin \varphi \\ y &= \cos \varphi \end{aligned}$$

$$T(e_2) = \begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix}$$

$$A = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$$